

Wirefree

Connectivity

Emerging wireless networking technologies are ready to cut you free

The PC is the centre of your digital universe, and it ties this universe together with the tangled cables of the information age. The mess of wiring behind your computer could all be gone, the cable sticking out of your PC Card Ethernet adapter could be unnecessary, the PDA cradle sitting on your desk could be counting sheep in your bottom drawer, and of course, you could be working while sipping on a Pina Colada by the seaside. All you need is a couple of little upgrades...to wirefree connectivity. Every single cable you see on your desk has a wireless equivalent just itching to replace it.

Zoning the world

Information connectivity can be conceptually divided into three zones, based upon the distance it needs to span. The shortest zone is the one immediately around you, your personal space. Let's assume you have a cell phone/PDA in one pocket, and an MP3 player hooked to your belt. Now before you hit the road you need to Hotsync your PDA and backup your phonebook and also download fresh music to your MP3 player. For this you first need to disentangle the cables and then connect the cables for each of your mobile devices. But, if you have a Personal Area Network (PAN) powered with say, Bluetooth, then all you need to do is simply point and click! A Wireless PAN is like a next-generation Infrared, far more sophisticated and far more capable, that creates a kind of 'data-island' over a short radius for your information. It will connect all your personally accessible devices, plus eliminate all wires from your desktop

for the printer or scanner or digital camera or any device that needs to exchange data with any other. Bluetooth is now being incorporated into many personal devices including top-range PDA models and cellular phones, as well as laptops.

Local Area Networks (LAN) with wires like the one you are plugged into right now continue to be the mainstay of networking infrastructure. Yet if you are the type who carries your computing device, a laptop or one of the newer PDAs around, you could walk into to your office and automatically get on the company network using IEEE802.11 wireless technology. With a range of up to 300 metres, LANs offer high-speed permanent connectivity to other 'data-islands'. Equipped with a PC Card for your laptop or a CF Card based Wireless LAN (WLAN) adapter for your PocketPC device, you'd be all set to network with your counterparts in any location within your building.

However, on the road you are unlikely to find WLAN connectivity as easily as the connectivity you could get over the

cell phone network. With the launch of digital GPRS services, data connectivity over long distances is much more convenient. Connections over long distances are the domain of Wide Area Networks (WAN) that connect far-flung devices, and even complete networks, to each other. What could be more convenient than a laptop that connects to the WLAN in the office as you walk in, and seamlessly switches to GPRS as you walk out, leaving you permanently connected no matter where you go?!

As business-oriented technologies, these three are the ones you can expect to use the most. But home networking is another ball-game altogether. At home, you have a wide range of devices that are all wired at some point, some communication oriented, some information/entertainment oriented, all with different needs. Yet there are many benefits to be

